

D496

Introduction to Data Science

Armored Drill

8 concepts · 12 items · 4 transfer items

What this is. Every concept in this course's public description, re-asked with more disguise than a practice test uses. Each stem carries facts that do not decide it. Every wrong answer is a named trap.

Source. Scope from the course's public description in the WGU Institutional Catalog (September 2026); items written from the field's standard body of knowledge. No WGU assessment item, course material or proprietary content is used or reproduced. Not affiliated with or endorsed by Western Governors University.

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How to run it

Paper first. Pencil. One pass, no notes, a timer if you want the pressure.

1. **Before you read the options**, fill the strip under the item: the target word (what is actually being asked), and the one fact in the stem that decides it.
2. **Principle before arithmetic**. Name the rule, then compute.
3. **Camouflage**. Most of every stem does not matter. Ask: which single fact decides this?
4. **Inverse pairs are the bait**. If two options are opposites, one of them is the trap built for you.
5. **Mark sure, unsure or guess** honestly. Sure-and-wrong is the most valuable miss on the page — it is a wrong mental model.
6. **Absolutes** (always, never, guarantee) are usually wrong. And recognizing an answer is not the same as owning it.
7. Items with a **heavy black bar** on the left are **transfer items**: the same principle, reversed or solved for a different variable.
8. Stop at the STOP page. Then grade, fill the triage grid, and tally the traps you fell for. The biggest tally is what you drill next.

The strip, filled in, looks like this:

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	_____	

Items

1.

A data scientist has 50,000 past taxi trips with pickup time, trip distance, fare, and tip. The file came from a city open data portal. Which question can this data set answer?

- A. Which city's open data portal is most reliable?
- B. Why do riders choose taxis over public transit?
- C. How satisfied are drivers with their pay?
- D. Do longer trips receive larger tips on average?

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

2.

A subscription app with 1.5 million users tests whether a new onboarding tutorial increases the 30-day retention rate. The design team built the tutorial in two sprints. Which is the correct null hypothesis?

- A. Retention for tutorial users equals 30%.
- B. The tutorial does not increase the 30-day retention rate.
- C. The tutorial will raise retention by exactly 5% for every user.
- D. The tutorial increases the 30-day retention rate.

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

3. [TRANSFER — reversed stem / different variable]

An analyst writes H1: the mean delivery time with new routing software is less than 45 minutes. What is H0?

- A. $\mu < 45$ minutes
- B. The sample mean equals 45 minutes.
- C. $\mu \geq 45$ minutes
- D. $\mu > 45$ minutes

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

4.

A test of a new ad at a sporting-goods chain with 80 stores returns $p = 0.02$. The ad was filmed at a local stadium. Which interpretation is correct?

- A. There is a 2% chance the null hypothesis is true.
- B. There is a 98% chance the ad works.
- C. If the ad truly had no effect, results this extreme would occur about 2% of the time.
- D. The ad's effect is large.

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

5. [TRANSFER — reversed stem / different variable]

Which result gives the strongest evidence against the null hypothesis?

- A. $p = 0.40$
- B. $p = 0.049$
- C. A sample of 10,000 with no p-value reported.
- D. $p = 0.001$

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

6.

A school district with 30 schools wants to know whether a tutoring app raises math scores. Principals may volunteer, and 12 do. The app vendor offers free licenses. What design gives the most trustworthy causal answer?

- A. Randomly assign schools (or students) to use the app or not, then compare scores.
- B. Measure scores after the app is used, with no comparison group.
- C. Use the app everywhere because the licenses are free.
- D. Compare the 12 volunteer schools with the other 18.

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

7.

An analyst estimates the average order value at an online florist as \$62, with a 95% confidence interval of \$58 to \$66. The analysis used 1,200 orders from spring. Which interpretation is correct?

- A. The method captures the true mean order value about 95% of the time; this sample's interval is \$58 to \$66.
- B. The true mean is exactly \$62.
- C. There is a 95% chance the sample mean is in the interval.
- D. 95% of individual orders fall between \$58 and \$66.

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

8. [TRANSFER — reversed stem / different variable]

A 95% confidence interval for mean wait time is 10 to 14 minutes. What is the margin of error?

- A. 12 minutes
- B. 4 minutes
- C. 24 minutes
- D. 2 minutes

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

9.

A data scientist's pandas script uses a Python for-loop over 5 million rows to multiply price by quantity. It takes 20 minutes on her company laptop, which is two years old. Which change most speeds it up?

- A. Replace the loop with one column operation: `df['total'] = df['price'] * df['qty']`.
- B. Buy a newer laptop.
- C. Nothing; loops are always as fast as vectorized code in pandas.
- D. Convert the DataFrame to a list of dictionaries and loop over that.

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

10.

A spreadsheet from a hospital network with 14 clinics has one row per clinic and separate columns for 2022, 2023, and 2024 visit counts. The file was emailed by the finance office. How should it be reshaped to be tidy?

- A. Ask finance to resend it in a different file type.
- B. Melt to long format: one row per clinic per year, with columns clinic, year, and visits.
- C. Pivot wider so each clinic becomes its own column.
- D. Drop the year columns and keep clinic names.

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

11.

An e-commerce test with 4 million visitors finds a new button raises conversion from 3.00% to 3.02%, with $p < 0.001$. Building the change costs \$250,000. The test was run by a new hire. What should the analyst report?

- A. Discard the result because a new hire ran the test.
- B. The result is not significant.
- C. The lift is statistically significant but very small; weigh the 0.02-point gain against the \$250,000 cost.
- D. The effect is large because p is so small.

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

12. [TRANSFER — reversed stem / different variable]

A study of 30 stores finds a promotion lifted sales by 12%, but $p = 0.09$ at $\alpha = 0.05$. What is the best summary?

- A. It is significant because 0.09 is below 0.10.
- B. The estimated effect is large but not statistically significant; more data is needed before concluding.
- C. The promotion is proven not to work.
- D. It is significant because a 12% lift is big.

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

STOP

Answers and dissection follow. Finish every item and every strip first.

Key and dissection

Struck-through text is the camouflage: facts that do not decide the item.

1. Answer D Concept 1

A data scientist has 50,000 past taxi trips with pickup time, trip distance, fare, and tip. ~~The file came from a city open data portal.~~ Which question can this data set answer?

Target word: answerable question **Decides:** Only time, distance, fare, and tip are recorded.

A	Camouflage	Answers the source detail, not the data.
B	Adjacent category	Needs survey data on motives, which the file lacks.
C	Dropped element	There are no driver or satisfaction fields.
D	Correct	Distance and tip are both in the data.

2. Answer B Concept 2

A subscription app ~~with 1.5 million users~~ tests whether a new onboarding tutorial increases the 30-day retention rate. ~~The design team built the tutorial in two sprints.~~ Which is the correct null hypothesis?

Target word: null hypothesis **Decides:** The null states no effect.

A	Wrong stem number	Mistakes the 30-day window for a 30% rate.
B	Correct	The null is the no-effect claim the test tries to reject.
C	Absolute language	A specific, certain claim, not a no-effect null.
D	Inverted operation	That is the alternative hypothesis.

3. Answer C Concept 2 · transfer

An analyst writes H1: the mean delivery time with new routing software is less than 45 minutes. What is H0?

Target word: null from the alternative — reverse **Decides:** The null takes the complement and holds the equals sign.

A	Confusion set	That restates H1.
B	Adjacent category	Hypotheses are about the population mean, not the sample.
C	Correct	The complement of $\mu < 45$, including equality.
D	Dropped element	Drops the equality case.

4. Answer C Concept 3

A test of a new ad at a sporting-goods chain ~~with 80 stores~~ returns $p = 0.02$. ~~The ad was filmed at a local stadium.~~ Which interpretation is correct?

Target word: p-value meaning **Decides:** Probability of data this extreme if the null is true.

A	Confusion set	A p-value is not the probability of the null.
B	Inverted operation	Subtracting p from 1 does not give the probability of the alternative.
C	Correct	That is the definition of a p-value.
D	Adjacent category	Significance says nothing about size.

5. Answer D Concept 3 · transfer

Which result gives the strongest evidence against the null hypothesis?

Target word: rank evidence — reverse **Decides:** The smaller the p-value, the stronger the evidence.

A	Inverted operation	Large p-values are weak evidence.
B	Confusion set	Just under 0.05 is significant but weaker than 0.001.
C	Dropped element	Sample size alone is not evidence of an effect.
D	Correct	The smallest p-value is the strongest evidence.

6. Answer A Concept 4

A school district with 30 schools wants to know whether a tutoring app raises math scores. Principals may volunteer, and 12 do. ~~The app vendor offers free licenses.~~ What design gives the most trustworthy causal answer?

Target word: causal design **Decides:** Volunteers self-select.

A	Correct	Randomization balances other differences between groups.
B	Dropped element	Without a control, the change cannot be credited to the app.
C	Camouflage	Cost does not answer whether it works.
D	Confusion set	Volunteer schools may differ in ways that affect scores.

7. Answer A Concept 5

An analyst estimates the average order value at an online florist as \$62, with a 95% confidence interval of \$58 to \$66. ~~The analysis used 1,200 orders from spring.~~ Which interpretation is correct?

Target word: CI interpretation **Decides:** The interval estimates the population mean.

A	Correct	Confidence describes the method's long-run capture rate.
B	Absolute language	\$62 is an estimate, not a certainty.
C	Adjacent category	The sample mean is the center; it is always inside.
D	Confusion set	The CI is about the mean, not individual orders.

8. Answer D Concept 5 · transfer

A 95% confidence interval for mean wait time is 10 to 14 minutes. What is the margin of error?

Target word: margin of error from the interval — reverse **Decides:** Margin = half the width.

A	Wrong stem number	The point estimate.
B	Dropped element	The full width, not half.
C	Inverted operation	Adds the endpoints instead of taking half the difference.
D	Correct	$(14 - 10) / 2 = 2$.

9. Answer A Concept 6

A data scientist's pandas script uses a Python for-loop over 5 million rows to multiply price by quantity. It takes 20 minutes on her company laptop, ~~which is two years old~~. Which change most speeds it up?

Target word: speed-up **Decides:** A row-by-row Python loop.

A	Correct	Vectorized operations run in optimized compiled code.
B	Camouflage	Answers the laptop's age; the loop remains the bottleneck.
C	Absolute language	Vectorized code is typically many times faster.
D	Inverted operation	Moves further from vectorized code.

10. Answer B Concept 7

A spreadsheet from a hospital network ~~with 14 clinics~~ has one row per clinic and separate columns for 2022, 2023, and 2024 visit counts. ~~The file was emailed by the finance office~~. How should it be reshaped to be tidy?

Target word: tidy reshape **Decides:** Year is a variable stored in headers.

A	Camouflage	The file type is not the problem; the layout is.
B	Correct	Year becomes a column and each row is one observation.
C	Inverted operation	Moves another variable into headers.
D	Dropped element	Throws away the measurements.

11. Answer C Concept 8

An e-commerce test with 4 million visitors finds a new button raises conversion from 3.00% to 3.02%, with $p < 0.001$. Building the change costs \$250,000. ~~The test was run by a new hire~~. What should the analyst report?

Target word: how to report **Decides:** A huge sample makes a tiny lift significant.

A	Camouflage	Who ran it does not decide the finding.
B	Inverted operation	$p < 0.001$ is significant.
C	Correct	Separates significance from practical value.
D	Confusion set	Confuses significance with effect size.

12. Answer B Concept 8 · transfer

A study of 30 stores finds a promotion lifted sales by 12%, but $p = 0.09$ at $\alpha = 0.05$. What is the best summary?

Target word: large effect, not significant — reverse case **Decides:** Big estimate, but $p > \alpha$.

A	Wrong stem number	Uses a cutoff other than the stated alpha of 0.05.
B	Correct	Reports both size and uncertainty.
C	Absolute language	Failing to reject is not proof of no effect.
D	Confusion set	Size does not decide significance.

Triage grid

Write each item number in the box it belongs to.

	Right	Wrong
Sure	Owned.	Concept inversion — top priority. A wrong mental model; it survives rewording.
Unsure / guess	Recognition, not mastery. Re-drill with the transfer.	Not learned yet.

Trap tally

For every miss, find the letter you chose in the key and tally its trap. The biggest number is the pattern to drill.

Trap	What it does	In this drill	Your misses
Dropped element	Leaves out a required term or step.	6	
Wrong stem number	A figure copied from the prompt that is not the answer.	3	
Adjacent category	The right idea in the wrong bucket.	4	
Inverted operation	Direction flipped: add/subtract, cause/effect, higher/lower.	7	
Absolute language	Always, never, guarantee — where judgment applies.	4	
Confusion set	Every option is real; the test is exact role-matching.	7	
Camouflage	Answers a stem fact that was never the question.	5	

Anchor rows — the concept list

One row per concept: the rule to lock in, the trap that catches people, and the tell in the question that signals it.

#	Concept	Rule	Trap	Tell
1	Answerable questions	A good analysis question can be answered with the variables actually in the data set.	Asking about motives, feelings, or groups the data never measured.	A list of the data set's columns followed by candidate questions.
2	Null vs. alternative hypothesis	The null states no effect or no difference (and holds the equals sign); the alternative states the effect being tested.	Writing the hoped-for effect as the null.	"Which is the null hypothesis?"
3	What a p-value means	A p-value is the probability of results at least this extreme if the null were true; it is not the probability that the null is true.	Reading $p = 0.02$ as "a 2% chance the null is true."	An interpretation question about a p-value.
4	Random assignment and control groups	Randomly assigning units to treatment and control is what supports a causal conclusion.	Comparing volunteers with non-volunteers.	Self-selected groups or no comparison group.
5	Confidence intervals	A 95% CI comes from a method that captures the true parameter in about 95% of samples; the margin of error is half its width.	Saying 95% of individual values fall in the interval.	An interval around a mean and an interpretation question.
6	Vectorized code	Whole-column (vectorized) operations in pandas and NumPy are far faster than Python loops over rows.	Buying faster hardware instead of removing the row loop.	A for-loop over millions of DataFrame rows.
7	Tidy data	Each variable is a column, each observation is a row, and each value is a cell.	Keeping values such as years as column headers.	Columns named 2022, 2023, 2024.
8	Statistical vs. practical significance	A result can be statistically significant but too small to matter, or large but not statistically significant; report both size and uncertainty.	Treating a tiny p-value as proof of a big effect.	A very large sample with a tiny difference, or a small sample with a big difference.

Course scope (public description): Introduction to Data Science introduces the data analysis process and common statistical techniques necessary for the analysis of data. Students will ask questions that can be solved with a given data set, set up experiments, use statistics and data wrangling to test hypotheses, find ways to speed up their data analysis code, make their data set easier to access, and communicate their findings.